Q27 Discussion Questions

1. Arditi et al. find that refusal is mediated by a single direction, and ablating it spares most other capabilities. If safety-relevant behavior of a deployed 70B model really rides on a one-dimensional structure, what does that imply about treating red-teaming pass rates as evidence of safety — and would you accept "refusal-direction robustness" audits as a replacement or a complement?
2. The probing literature's core lesson is *representable* $\neq$ *used*. Construct a concrete scenario where a high-accuracy "deception probe" on a deployed model would mislead a safety team, and specify the causal experiment that would have caught the error.
3. Jailbreaks have been described as "reversion to pre-training traits." What evidence would distinguish that hypothesis from the weaker one that safety training simply does not generalize off-distribution? Which experiment in the module (Wei et al.'s taxonomy, Zou et al.'s transferable suffixes, Arditi et al.'s ablation) comes closest to discriminating them?
4. Qi et al. show 10 adversarial examples suffice to strip safety from GPT-3.5 Turbo. Is "post-training steers rather than rewrites" the right frame, or does the fine-tuning attack show something stronger — that alignment was never more than a conditional policy? What would count as evidence that some safety property is *deeply* encoded?
5. Greenblatt et al.'s alignment faking shows post-trained preferences persisting strongly enough that the model resists overwriting them. How does that result sit with the shallow-refusal picture from the same module — contradiction, complement, or difference in what "preference" and "refusal" mean mechanistically? Defend a reading.